AZIF NIZAM

Embedded Software Engineer

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CAREER OBJECTIVE

Embedded Software Engineer with strong proficiency in C and C++, and a solid foundation in system-level programming and embedded development. Hands-on experience in Linux system programming, socket-based communication, memory management, file handling, and low-level debugging. Interested in contributing to reliable and efficient embedded and system software solutions.

SKILLS

- **Programming**: C, C++, Python, Data structures
- **Operating System and System Programming**:
  - POSIX system calls (fork, exec, wait, pipe, signals)
  - Process management and IPC mechanisms
  - TCP/IP and UDP socket programming
  - File systems and low-level file I/O
  - Multi-threaded programming using pthreads
- **Embedded Systems:**
  - Microcontroller: ESP32, PIC (18F4580) board
  - Platform: Linux (ubuntu)
  - Processor architecture basics
  - Communication Protocols: UART, SPI, CAN, I2C
  - Peripherals: Timers, counters, interrupts
- **Development Tools:**
  - Compilers: GCC, G++, XC8 (cross compiler)
  - Version control: Git, GitHub
  - Debugger: GDB

EDUCATION

Advanced Embedded Systems Course (Ongoing) *June 2025 – March 2026*
(Linux Internals, Microcontrollers, RTOS)
Emertxe Information Technologies, Bengaluru, Karnataka

Bachelor of Technology (Electronics and Communication) *2021 - 2025*
Amal Jyothi College of Engineering (Autonomous), Kottayam, Kerala **CGPA 7.58/10**
(Affiliated to APJ Abdul Kalam Technological University)

Class 12th - Central Board of Secondary Education *2020 – 2021*
Bhavans vidya mandir, Pathanamthitta, Kerala **CGPA 7.4/10**

Class 10th grade- Central Board of Secondary Education *2018 – 2019*
Bhavans vidya mandir, Pathanamthitta, Kerala **CGPA 9.02/10**

PROJECTS

TFTP (Trivial File Transfer Protocol) Implementation *February 2026*

- Developed a UDP-based file transfer system in C implementing a simplified TFTP protocol with client–server architecture, supporting PUT and GET operations, block-wise data transfer (512-byte blocks), stop-and-wait ARQ mechanism, multi-mode transfer (default, octet, netascii), block numbering, ACK handling, retransmission logic, and file I/O using low-level system calls (open, read, write, lseek, close).

Mini Shell *January 2026*

- Developed a Unix-like mini shell in C supporting built-in and external command execution, foreground/background processing, job control (jobs, fg, bg), signal handling (SIGINT, SIGTSTP, SIGCHLD), and multi-stage pipelines using fork, execvp, pipe, and waitpid.

Inverted Search Engine *December 2025*

- Developed an inverted search engine in C using hash tables and linked lists to index and retrieve words across multiple text files. Implemented word normalization, collision resolution via chaining, file-wise frequency tracking, and a modular architecture to ensure fast, accurate, and scalable search operations.

Banking System (C++) *December 2025*

- Developed a console-based banking system in C++ using object-oriented programming principles including encapsulation, inheritance, and polymorphism. Implemented role-based authentication for bankers and customers, account management features such as customer creation, deposits, withdrawals, and transaction history, and a modular class-based architecture with file handling to ensure data integrity, scalability, and maintainability.

Arbitrary Precision Calculator *November 2025*

- Developed a C-based arbitrary-precision calculator supporting addition, subtraction, multiplication, and division on extremely large integers using custom doubly linked lists. Implemented digit-wise parsing, list-based arithmetic algorithms, and a modular program architecture to ensure accuracy, performance, and scalability.

Mp3 Tag Reader for ID3v2.3 *November 2025*

- Developed a command-line MP3 tag editor in C to view and modify ID3 metadata including title, artist, album, year, genre, and comments. Implemented binary file handling, structured data parsing, and modular program design to ensure maintainability, scalability, and reliable metadata manipulation.

Lexical Analyzer *October 2025*

- Developed a C-based lexical analyzer to scan and tokenize C source files, identifying keywords, identifiers, operators, constants, and special symbols. Implemented character-level parsing, modular architecture, efficient file I/O, and robust error detection for malformed tokens, demonstrating strong fundamentals in compiler design and system-level programming.

LSB Image Steganography *October 2025*

- Designed and implemented a C-based steganography tool using Least Significant Bit (LSB) techniques to hide and retrieve data within BMP images. Applied bitwise operations, structured file processing, modular design, and robust error handling through a command-line driven interface.

ACADEMIC PROJECTS (Btech)

L-S band Power Amplifier Integrated Antenna *June 2024 – March 2025*

- Designed a Class-AB RF power amplifier IC with output impedance matching to a microstrip patch antenna, enabling efficient signal transfer and integration for a portable electromagnetic imaging system.
- Applied RF design principles, impedance matching techniques, and antenna interfacing. Gained hands-on exposure to RF front-end design, radio architecture concepts, and RF performance considerations relevant to modem systems.
  Tool used to design: Keysight Advanced Design System (ADS) software

Automated Traffic Alert System (ATAS) *June 2023 – March 2024*

- Built an ESP32-based V2V communication prototype to broadcast and receive emergency vehicle alerts, integrating wireless antennas and real-time embedded firmware for on-road safety applications.
- Implemented embedded firmware interfacing with wireless modules using UART communication, gaining exposure to real-time communication systems and RF-enabled embedded software.
  Tool used to develop: Arduino IDE

INTERNSHIP AND WORKSHOP

- **Internship** (2023): Revertech IT Solutions-PCB Designing
- **Workshop** (2024): VLSI for beginners by NIELIT Ministry of Electronics and IT, Government of India
- **Workshop** (2023): Foundation of Electric and Hybrid Vehicles by Techmaghi and Tryst'24 IIT Delhi

CERTIFICATION

- **NPTEL** (April 2024) - A brief introduction of micro sensors, IISER, Bhopal

ACHIEVEMENTS

- **Young Innovators Program Kerala Development & Innovation Strategic Council** (2023): Certificate of recognition